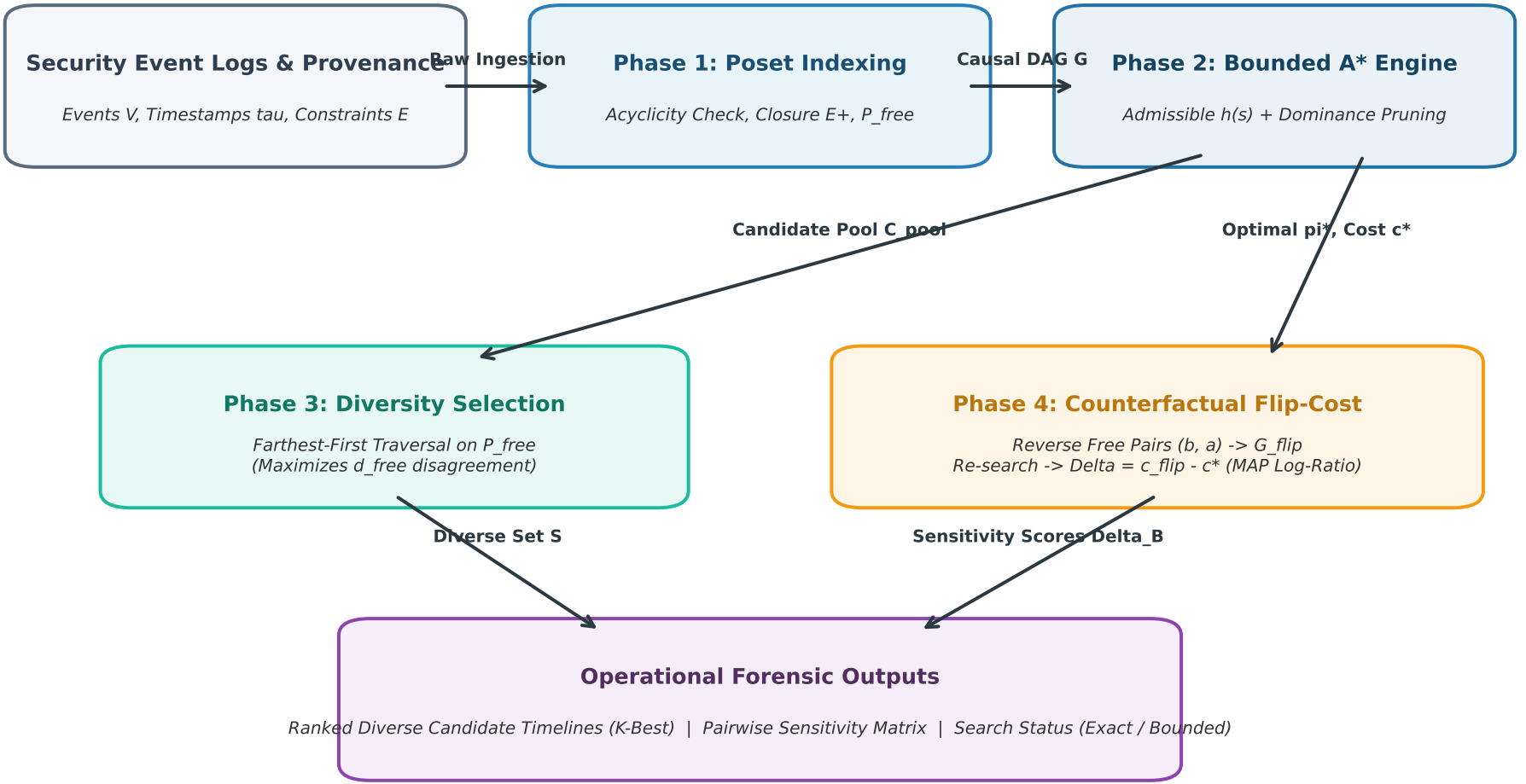
Security Event Logs & Provenance
Events V, Timestamps tau, Constraints E
Raw Ingestion
Phase 1: Poset Indexing
Acyclicity Check, Closure E+, P_free
Causal DAG G
Phase 2: Bounded A* Engine
Admissible h(s) + Dominance Pruning
Candidate Pool C_pool
Optimal pi*, Cost c*
Phase 3: Diversity Selection
Farthest-First Traversal on P_free
(Maximizes d_free disagreement)
Phase 4: Counterfactual Flip-Cost
Reverse Free Pairs (b, a) -> G_flip
Re-search -> Delta = c_flip - c* (MAP Log-Ratio)
Diverse Set S
Sensitivity Scores Delta_B
Operational Forensic Outputs
Ranked Diverse Candidate Timelines (K-Best) | Pairwise Sensitivity Matrix | Search Status (Exact / Bounded)